States in descending order with respect to number of large dams constructed here- Madhya Pradesh, Gujarat, Rajasthan, telangana, Andhra Pradesh. Significantly, the biggest dams in the country are built in Maharashtra, Madhya Pradesh and Gujarat respectively.

70. Consider the following pairs :

	Reservoirs		**States**
1.	**Ghataprabha**	**-**	**Telangana**
2.	**Gandhi Sagar**	**-**	**Madhya Pradesh**
3.	**Indira Sagar**	**-**	**Andhra Pradesh**
4.	**Maithon**	**-**	**Chhattisgarh**

How many pairs given above are not correctly matched?

(a) Only one pair (b) Only two pairs

(c) Only three pairs (d) All four pairs

I.A.S. (Pre) 2022

Ans. (c)

The correct match is -

(Reservoirs)	**(States)**
Ghataprabha	Karnataka
Gandhi Sagar	Madhya Pradesh
Indira Sagar	Madhya Pradesh
Maithon	Jharkhand

The raja Laxmagowda Dam (Hidkal Dam) located on the Ghataprabha River in Karnataka. Indira Sagar Dam is located on the Narmada River in Khandwa, Madhya Pradesh. Maithon Dam is on located Barakar River in Dhanbad, Jharkhand. So only three pairs are not matched.

71. Which hydropower plant in Bhutan was inaugurated recently by Indian Prime Minister Narendra Modi?

(a) Chhukha Power Plant (b) Dagachhu Power Plant

(c) Kurichha Power Plant (d) Mangdechhu Power Plant

U.P.P.C.S. (Pre) 2019

Ans. (d)

In 2019, Indian Prime Minister, Narendra Modi inaugurated Mangdechhu Hydropower Plant in Bhutan.

Agriculture

*The agriculture and allied sector constitute the foundational pillar of the Indian economy, engaging over 50% of the workforce and contributing approximately 18% to the country's Gross Value Added (GVA). According to Economic Survey 2024-25, this sector contributes approximately 16% of the country's GDP for FY, 2024(PE) at current prices and supports about 46.1% of the population. Not only does its performance directly impact food security, but it also influences other sectors, sustaining livelihoods and supporting economic growth.

*First Agricultural University of the country was established in 1960. Inauguration of this University **(Uttar Pradesh Agricultural University)** was done by Jawaharlal Nehru on 17 November 1960 in **Pantnagar**. Later, it was named as **Govind Ballabh Pant University of Agriculture and Technology.** *There are **15 Agro-Climate Zones** in India.

S.N.	**Agro-Climate Zone**	**State/U.T.**
1.	Western Himalayan Region	Ladakh, Jammu & Kashmir, Himachal Pradesh, Uttarakhand
2.	Eastern Himalayan Region	Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura, West Bengal
3.	Lower Gangetic Plains Region	West Bengal
4.	Middle Gangetic Plains	Uttar Pradesh, Bihar
5.	Upper Gangetic Plains	Uttar Pradesh
6.	Trans - Gangetic Plains	Chandigarh, Delhi, Haryana, Punjab, Rajasthan
7.	Eastern Plateau & Hills	Madhya Pradesh, Chhattisgarh, Jharkhand, Odisha, West Bengal, Maharashtra
8.	Central Plateau & Hills	Madhya Pradesh, Rajasthan, Uttar Pradesh
9.	Western Plateau & Hills	Madhya Pradesh, Maharashtra
10.	Southern Plateau & Hills	Andhra Pradesh, Telangana, Karnataka, Tamil Nadu
11.	East Coast Plains & Hills	Andhra Pradesh, Odisha, Puducherry, Tamil Nadu
12.	West Coast Plains & Hills	Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu
13.	Gujarat Plains & Hills	Gujarat, Dadra and Nagar Haveli and Daman & Diu
14.	Western Dry Zone	Rajasthan
15.	Islands zone	Andaman and Nicobar Islands, Lakshadweep

According to the National Bureau of Soil Survey, "India is divided into **20 Agro-Ecological Region** and **60 Agro-Ecological Sub-regions"**. It is classified on the basis of Soil, types of climate and natural geographical conditions.

Agro - Ecological Regions	
Ecosystem	Agro- Ecological Region
Arid Ecosystem	1. Western Himalayas 2. Western Plain, Kachchh and part of Kathiwara Peninsula 3. Deccan Plateau
Semi Arid Ecosystem	4. Northern Plains (Upper Ganga Plain) 5. Northern Plains (Rajasthan High lands and Gujarat Plain) 6. Northern Plains (Middle Ganga Plain) 7. Deccan Plateau (Malwa Plateau, Gujarat Plain and Kathiawar Peninsula) 8. Deccan Plateau (Mixed red and Black soil.) 9. Deccan Plateau (Red Sandy Soil)
Subhumid Ecosystem	10. Eastern Plateau (Satpura Range and Mahanadi Basin) 11. Eastern Plateau (Bundelkhand High lands) 12. Eastern Plateau (Red and Laterite Soil) 13. Northern Plains (Lower Ganga Plains). 14. Western Himalayas (Jammu & Kashmir warm region, Himachal Pradesh, Uttarakhand) 15. Bengal Basin
Humid-Perhumid Ecosystem	16. Assam and North Bengal Plains 17. Eastern Himalayas 18 Purvanchal Hills (North-Eastern Hills)
Coastal Ecosystem	19. Eastern Coastal Plains and Andaman and Nicobar Islands 20. Western Ghats (Coastal Plains and Western Hills)

*'**History of Indian Agriculture**' is written by M.S. Randhava (Mohinder Singh Randhawa). According to standards of Food and Agriculture Organisation (FAO), for secure storage of food grains, relative humidity should be kept at 14%.

***Double cropping** means when two or more than two crops are grown in one crop year at one place. Intercropping is the practice of growing two or more crops in proximity. ***Mixed farming** refers to growing food and fodder crops and rearing livestock. By practicing this, farmers can increase their income.

*Punjab is leading state in implementing **'Contract farming'.** Sikkim is a hilly state in North-East. Its maximum area is covered under forest. Less than 10% of total land is available for agriculture in this state. On the contrary, Uttar Pradesh, Punjab, Haryana are the major grain-producing states. According to Economic Survey 2024-25, the state of Uttar Pradesh has the largest share in the total food grain production (2023-24) in India. As per the latest data 2021-22 (P) of the Ministry of Agriculture of India, Net Area Sown in India is 46.01%, Forest area is 23.49% and land under other practices is 30.5%. Among the top 10 rice producing countries in the world, India is the first and China is at the second place (FAO, 2023). About 36.18% of the gross area of food grain in India (2022-23) is under rice cultivation.

*On the basis of the data of the year 2021-22, the largest fertilizer consumption in terms of thousand tonnes in five states respectively Uttar Pradesh (5169.07 thousand tonnes), Maharashtra (3135.57 thousand tonnes), Madhya Pradesh (2651.73 thousand tonnes), Karnataka (2192.38 thousand tonnes) and Punjab (1989.76 thousand tonnes).

*Balanced fertilizers are used to increase production, to improve the quality of food grains and to maintain land productivity. ***Seed village concept** is a concept based on a village where a trained group of farmers are involved in production of seeds of various crops and cater to the needs of themselves and fellow farmers of the villages and farmers of the neighbouring villages in appropriate timely manner and at an affordable cost.

***Agmark** is a quality certification mark granted by the government of India under Agricultural produce (Grading and Marking) Act, 1937. Various steps have been taken by the Government of India for the continuous development of agriculture. Enrichment of sustainable soil fertility through soil health card scheme, increased capacity of water through the **Pradhan Mantri Gram Sinchayi Yojna** and improving access to irrigation, organic farming has been provided support through Paramparagat Krishi Vikas Yojna (PKVY) and also several steps have been taken for the creation of **'Integrated National Agriculture Market'** so that the farmers income may increase.

***Kisan Credit Card (K.C.C.)** scheme is operating throughout the country with the help of commercial banks, co-operative banks and regional-rural banks.

Green Manure: Prior to the sowing of the crop seeds, some plants like sun hemp or guar are grown and then mulched by ploughing them into the soil. These green plants thus turn into green manure which helps in enriching the soil in nitrogen and phosphorus.

*The nitrogen level available in Dhaincha is (0.42%), 0.43% in sanai and 0.34% in Guar but by sowing sun hemp or Sanayi, the land gets the maximum amount of Nitrogen 86-129 kg/Hectare. A farm gets 84-105 kg/hectare nitrogen from Dhaincha, 68-85 kg/hectare from Guar and 74-88 kg/hectare from Cobia.

1. How is permaculture farming different from conventional chemical farming?

1. Permaculture farming discourages monocultural practices but in conventional chemical farming, monoculture practices are pre-dominant.

2. Conventional chemical farming can cause increase in soil salinity but the occurrence of such phenomenon is not observed in permaculture farming.

3. Conventional chemical farming is easily possible in semi-arid regions but permaculture farming is not so easily possible in such regions.

4. Practice of mulching is very important in permaculture farming but not necessarily so in conventional chemical farming.

Select the correct answer using the code given below:

(a) 1 and 3 (b) 1, 2 and 4

(c) 4 only (d) 2 and 3

I.A.S. (Pre) 2021

Ans. (b)

The method of agriculture that makes more use of the resources of nature without misusing or polluting it can be called sustainable agriculture. In this, agriculture is done by adopting soil water and naturally made manure and for its conservation, crop rotation, covering the soil under mulching method, etc. So statement 1, 2, 4 are true. This method is effective even in dry semi-arid regions, so statements 3 is false.

2. What is/are the advantage/advantages of zero tillage in agriculture?

1. Sowing of wheat is possible without burning the residue of previous crop.

2. Without the need for nursery of rice saplings, direct planting of paddy seeds in the wet soil is possible.

3. Carbon sequestration in the soil is possible.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only

(c) 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2020

Ans. (d)

Zero tillage is the process where the crop seed will be sown directly through drillers without prior land preparation and thus avoiding disturbing the soil where previous crop stubbles are present.

Zero tillage and direct seeded rice (DSR) not only enable sowing of wheat without any burning of crop residue, but also save water by doing away with transplanting operations in paddy.

Zero tillage is environmentally safe reducing greenhouse effect by way of carbon sequestration.

3. What are the advantages of fertigation in agriculture?

1. Controlling the alkalinity of irrigation water is possible.

2. Efficient application of Rock Phosphate and all other phosphatic fertilizers is possible.

3. Increased availability of nutrients to plants is posible.

4. Reduction in the leaching of chemical nutrients is possible.

Select the correct answer using the code given below:

(a) 1, 2 and 3 only (b) 1, 2 and 4 only

(c) 1, 3 and 4 only (d) 2, 3 and 4 only

I.A.S. (Pre) 2020

Ans. (c)

Fertigation is a method of fertilizer application in which fertilizer is dissolved into the irrigation water through the drip system.

Urea, potash and highly water soluble fertilizers are available for use through fertigation. Use of super phosphorus through fertigation must be avoided as it results in precipitation of phosphate salts. Thus, second statement is not correct and the correct answer is (c).

4. In the context of India, which of the following is/are considered to be practice(s) of eco-friendly agriculture?

1. Crop diversification

2. Legume intensification

3. Tensiometer use

4. Vertical farming

Select the correct answer using the code given below:

(a) 1, 2 and 3 only (b) 3 only

(c) 4 only (d) 1, 2, 3 and 4

I.A.S. (Pre) 2020

Ans. (d)

Crop diversification : It refers to the adoption of diverse new crops to the existing cropping pattern, consistent with the ago-climatic conditions of the area besides the ground water conditions. It helps in moving away from nonculture and water-guzzling crops characterinzing agriculture practices in India.
Legume (they fix the atmospheric nitrogen) intensification is an eco-friendly agricultural practice.
Tensiometers (a sealed, water-filled tube with a ceramic porous cup and a vaccum gauge at the top) could be really helpful in providing estimates of soil moisture, thus helping in water conservation.
Vertical farming : Crops are grown indoors, under artificial conditions of light and temperature. It uses significantly reduced water and pesticides than in traditional agriculture.

5. Who has written 'The History of Indian Agriculture?

(a) M. S. Swaminathan (b) S. Ayyapan
(c) K.B. Thomas (d) M.S. Randhawa

U.P.P.C.S. (Mains) 2015

Ans. (d)

'The History of Indian Agriculture' is written by M.S. Randhawa. His full name is Mohinder Singh Randhawa. He played a major role in the field of agriculture research and Green Revolution in India.

6. Read the following statements and choose the correct option :

Statement I : India has been divided into 20 agro-climatic regions
Statement II : India has been divided into 15 agro-ecological regions
Statement III : Coverage area of Western Himalaya cold arid ecoregiona is more than coverage area of Western Himalaya region

(a) Statement I, II and III all are correct
(b) Statement I, II and III all are incorrect
(c) Only Statement I and II are correct
(d) Only Statement I is correct
(e) None of the above / More than one of the above

Chhattisgarh P.C.S. (Pre) 2020

Ans. (b)

For resource development, the country has been divided into 15 agro-climatic regions.
There are 20 agro-ecological regions according to the National Bureau of Soil Survey & Land use Planning.
Coverage area of Western Himalaya cold arid eco region is more than coverage area of Western Himalaya region.

7. Consider the following statements with reference to Upper Gangetic agro-climatic region:

1. This region encompasses central and western Uttar Pradesh.
2. This region has more than 130% irrigation intensity and 140% cropping intensity.

Which of the above statements is/are correct?

(a) Both 1 and 2 (b) Neither 1 nor 2
(c) Only 1 (d) Only 2

UPPSC (Pre) 2024

Ans. (a)

The central and western parts of Uttar Pradesh are included in the Upper Ganga Plain agro climatic zone. This region comprises four Agro-climatic zone, the central - western plain zone, the south western semi - Arid zone, western plain zone and the central plain zone. The irrigation intensity in this region is approximately 131% (≈approx 130%) and the agricultural/cropping intensity is about 142% (≈approx 140%). Therefore, both statements are correct.

8. The agro-ecological regions of the country are –

(a) 15 (b) 17
(c) 18 (d) 20

U.P.R.O./A.R.O. (Mains) 2013
U.P.R.O./A.R.O. (Spl) (Pre) 2010

Ans. (d)

According to the National Bureau of Soil Survey, India has been divided into 20 Agro-Ecological Zones (AEZs). Each AEZ is as uniform as possible in terms of physiography, climate, length of growing period and soil type for macro level land-use planning and effective transfer of technology.

9. The total number of Agro-ecological zones in India is –

(a) 15 (b) 17
(c) 19 (d) 20

U.P.P.C.S. (Mains) 2016

Ans. (d)

India is divided into 20 Agro-Ecological Zone. Each Agro-ecological Zone is uniform in term of physiography, climate, length of growing period and soil type for macro-level land use planning and effective transfer of technology.

10. P. Sengupta and G. Sdasyuk (1968) had divided India into how many micro agricultural regions?

(a) 58 (b) 63
(c) 60 (d) 65

U.P. P.C.S. (Mains) 2017

Ans. (c)

The Registrar General of Census, Government of India published a monograph in 1968 based on the study of Dr. P. Sengupta and Russian geographer Dr. Galina Sdasyuk, in which India was divided into 60 micro agricultural regions.

11. Assertion (A) : The dry zone of India has a predominantly agrarian economy.

Reason (R) : It has large potential for the second Green Revolution.

Select the correct answer from the code given below:

Code :

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

U.P.P.C.S. (Mains) 2010

Ans. (b)

Green Revolution can be brought in dry zones of India by providing irrigation facility. Agriculture is the base of the economy of these areas. Thus, Assertion (A) and Reason (R), both are true however, Reason (R) does not explain the Assertion (A).

12. In the context of food and nutritional security of India, enhancing the 'Seed Replacement Rates' of various crops helps in achieving the food production targets of the future. But what is/are the constraint/ constraints in its wider/greater implementation?

1. There is no National Seeds Policy in place.

2. There is no participation of private sector seed companies in the supply of quality seeds of vegetables and planting materials of horticultural crops.

3. There is a demand-supply gap regarding quality seeds in case of low value and high volume crops.

Select the correct answer using the code given below.

(a) 1 and 2 (b) 3 only

(c) 2 and 3 (d) None

I.A.S. (Pre) 2014

Ans. (b)

The National Seed Policy 2002 is in place in India, so the statement (1) is false. It is not true that there is no participation of private sector seed companies in the supply of quality seeds of vegetables and planting materials of horticultural crops, but they mainly produce high-priced seeds in lower volume. So it is only available to few farmers. They supply nearly the entire hybrid seeds required for vegetables. There is demand-supply gap regarding quality seeds in case of low value and high volume crops, Thus, only statement (3) is correct.

13. The first Agricultural University in the country was set up in the year –

(a) 1950 (b) 1960

(c) 1970 (d) 1980

U.P.P.C.S. (Mains) 2013

Ans. (b)

Govind Vallabh Pant University of Agriculture and Technology is the first agricultural university of India. It was inaugurated by Jawaharlal Nehru on 17 November 1960 as the "Uttar Pradesh Agricultural University" (UPAU) in Pantnagar, Uttarakhand (Earlier it was the part of Uttar Pradesh). Later the name was changed to "Govind Ballabh Pant University of Agriculture and Technology" in the memory of the first Chief Minister of Uttar Pradesh.

14. The first Agricultural University of the country is

(a) J.N.K.V., Jabalpur (b) G.B.P.A.U., Pant Nagar

(c) P.A.U., Ludhiana (d) R.A.U., Bikaner

U.P.P.C.S. (Mains) 2014

Ans. (b)

See the explanation of the above question.

15. First Agriculture University in India was established in the year –

(a) 1955 (b) 1960

(c) 1965 (d) 1970

U.P.R.O./A.R.O. (Mains) 2013

Ans. (b)

See the explanation of the above question.

16. If safe storage is to be ensured, the moisture content of food grains at the time of harvesting should NOT be higher than

(a) 14% (b) 16%

(c) 18% (d) 20%

I.A.S. (Pre) 1994

Ans. (a)

According to the Food and Agriculture Organization (FAO) standards, if safe storage is to be ensured, the moisture content of food grains at the time of harvesting should not be higher than 14%.

17. The approximate representation of land use classification in India is –

(a) Net area sown 25%; forests 33%; other areas 42%

(b) Net area sown 58%; forests 17%; other areas 25%

(c) Net area sown 43%; forests 29%; other areas 28%
(d) Net area sown 47%; forests 23%; other areas 30%

I.A.S. (Pre) 2010

Ans. (d)

According to the data for the year 2009-10, Net area sown is 45.27%, forests 23.28% and other areas is 31.45% . So, the closest answer is (d). According to data of 2021-22 (P) released by Ministry of Agriculture- Net Area Sown is 46.01%, Forest Area 23.49%, and others are 30.5%.

18. Double cropping in agriculture means raising of
(a) Two crops at different times
(b) Two crops simultaneously
(c) One crop along with another crop
(d) None of these

43rd B.P.S.C. (Pre) 1999

Ans. (a)

The practice of consecutively producing two crops of either like or unlike commodities on the same land within the same year is called double cropping. An example of double cropping might be to harvest a wheat crop by early summer and then plant corn or soyabeans on that acreage for harvest in the fall. This practice is only possible in regions with long growing seasons. While the cultivation of two or more crops simultaneously on the same field is called intercropping.

19. Which of the following is the chief characteristic of 'mixed farming'?
(a) Cultivation of both cash crops and food crops
(b) Cultivation of two or more crops in the same field
(c) Rearing of animals and cultivation of crops together
(d) None of the above

I.A.S. (Pre) 2012

Ans. (c)

Mixed farming has equal emphasis on crop cultivation and animal husbandry. Crop rotation and intercropping play an important role in maintaining soil fertility. Animals like sheep, cattle pigs and poultry provide the main income along with the crops.

20. Mixed farming consists of:
(a) Growing of several crops in a planned way
(b) Growing rabi as well as Kharif crops
(c) Growing several crops and also rearing animals
(d) Growing of fruits as well as vegetables

R.A.S./R.T.S.(Pre) 2010

Ans. (c)

See the explanation of the above question.

21. Which one is an example of "Parallel Cropping"?
(a) Potato + Rice (b) Wheat + Mustard
(c) Cotton + Wheat (d) Sorghum + Potato

Uttarakhand P.C.S. (Pre) 2016

Ans. (b)

Parallel cropping is an agricultural method by which two crops are grown together, in the same land in parallel rows. Significantlyh, both crops grown do not compete for nutrients and have no effect on each other. The correct answer to the question is option (b). Wheat and mustard are examples of parallel crops.

22. In the given states leaving......... percentage of agriculture land is excessive.
(a) Punjab (b) Haryana
(c) Uttar Pradesh (d) Sikkim

Uttarakhand P.C.S. (Pre) 2006

Ans. (d)

Sikkim is a hilly state in Northeast India. Most of its parts are covered by forest. Out of the total land, less than 10% part is available for agriculture.

23. Which of the following is not a characteristic of Indian Agriculture?
(a) Over-dependence on nature
(b) Low level of productivity
(c) Diversity of crops
(d) Predominance of large farms

Uttarakhand P.C.S. (Pre) 2010

Ans. (d)

In India, there is a dominance of small and marginal holdings rather than large farms. Rest are the characteristics of Indian agriculture.

24. The reasons for low productivity in Indian agriculture is-
(a) Overcrowding in Agriculture
(b) Small Land Holding
(c) Traditional agricultural practices
(d) All of the Above

U.P.P.C.S. (Pre) 2007

Ans. (d)

Overcrowding in agriculture, small land holding and traditional agricultural practices all three are among the main reasons for low productivity in Indian agriculture.

25. Assertion (A) : Green Revolution Technology played a crucial role in gradually transforming traditional agriculture into a modern scientific one.

Reason (R) : It did not involve much of social and environmental costs.

Select the correct answer from the codes given below:

Code :

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is false

(d) (A) is false, but (R) is true

U.P.P.C.S. (Mains) 2014

Ans. (b)

Green Revolution Technology played a crucial role in gradually transforming traditional agriculture into modern scientific agriculture with the use of irrigation, specialized seeds, fertilizers, pesticides and machinery. So, Assertion (A) is true. Green Revolution does not involve social and environmental cost. Its success was due to a combination of high rates of investment in crop research, infrastructure, etc. The environmental cost is widely recognised as a potential threat for long term sustainability. Thus, (R) is also true but (R) is not the correct explanation of (A).

26. Which of the following is not the reason for low productivity in Indian agriculture?

(a) Pressure of the population on the land

(b) Disguised unemployment

(c) Cooperative farming

(d) Small Land Holding

U.P. P.C.S. (Pre) 2003

Ans. (c)

The pressure of the population on the land, disguised unemployment and small land holding are the reasons for low productivity in Indian agriculture whereas, cooperative farming is one of the means of the development of agriculture. Cooperative farming is farming where each member farmer is the owner of his land, but farming is done jointly and profit is distributed among member farmers.

27. The average size of operational holdings in India is the largest in –

(a) Punjab (b) Gujarat

(c) Madhya Pradesh (d) Rajasthan

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (d)

The average size of operational holdings (Agriculture Census, 2015-16) in India is largest in Rajasthan followed by Maharashtra and Uttar Pradesh.

28. Agriculture in India is considered as :

(a) A means of livelihood (b) A profession

(c) A trade (d) An industry

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (a)

Agriculture in India is considered as a means of livelihood. According to official, maximum man power of India is engaged in agriculture and related areas but contribute only a small portion to the total GDP.

29. With reference to Indian agriculture, which one of the following statements is correct?

(a) About 90 percent of the area under pulses in India is rainfed

(b) The share of pulses in the gross cropped area at the national level has doubled in the last two decades

(c) India accounts for about 15 percent of the total area under rice cultivation in the world

(d) Rice occupies about 34 percent of the gross cropped area of India.

I.A.S. (Pre) 2002

Ans. (a)

At the time of this question, only 12.7% (in 2011-12 it was 16.1%) area under Pulses agriculture had facilities of irrigation. Rest of the area under pulses cultivation is rainfed. So, statement (1) is almost correct.

If we look at the agriculture of two decades in 1990-91 the area under pulses agriculture was 246.62 Lakh hectare whereas it is reduced to 227.6 Lakh hectare in 2004-05 (27.5 Million Hectare in 2023-24. So statement (2) is not correct which says that share of pulses in the gross cropped area has doubled in the last two decades.

India accounts for about 29% of the total area under Rice cultivation in the world. So statement (3) is also wrong. Rice occupies about 23.7% (22.33% in 2011-12) of the gross cropped area of India, not 34%. Thus, statement (4) is also wrong. About 36.18% of the gross area of food grain in India (2022-23) is under rice cultivation.

30. The two largest consumers of chemical fertilizers in India are

(a) Andhra Pradesh and Maharashtra

(b) Punjab and Haryana

(c) Punjab and Uttar Pradesh

(d) Uttar Pradesh and Andhra Pradesh

U.P.P.C.S. (Mains) 2009

Ans. (d)

According to data for year 2013-14, the fertilizer consumption of the given States are –

State	Total fertilizer consumption (Thousand tons)
Uttar Pradesh	3842.04
Andhra Pradesh	3119.43

At present 2021-22, the Three largest consumers of chemical fertilizers are as follows –
Uttar Pradesh : 5169.07 (thousand tonnes) and Maharashtra (3135.57 thousand tonnes) & Madhya Pradesh (2651.73 thousand tonnes).

31. Which of the following green manure crops contains highest amount of nitrogen?

(a) Dhaincha (b) Sunhemp
(c) Cow pea (d) Guar

U.P.P.C.S. (Mains) 2016

Ans. (c)

Cow pea contains the highest amount of Nitrogen (0.49%) among the given green manures. Sunhemp (0.43%), Dhaincha (0.42%) and Guar has (0.34%), nitrogen content.

32. In newly improved arid land the crop suitable for the green manure is-

(a) Lobia (b) Dhaincha
(c) Green Gram(Moong) (d) Brown Hemp(Sanai)

U.P.P.C.S. (Mains) 2008

Ans. (a)

In newly improved arid land Lobia is a suitable crop for the green manure.

33. Balanced fertilizers are used to-

(a) Increase the production
(b) Improve fertilizer use efficiency
(c) Maintain the productivity of the soil
(d) All of these

U.P.P.C.S. (Mains) 2008

Ans. (d)

Balanced fertilizers are used to increase the production, improve quality of food grains and maintain the productivity of the soil.

34. In southern India, the area of high agricultural productivity is found in –

(a) Kerala coast (b) Tamil Nadu coast
(c) Telangana (d) Vidarbha

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (b)

The high agriculture productivity areas in southern India are coastal areas of Andhra Pradesh, Tamil Nadu, Area of Surat in Gujarat and areas of Satara and Kolhapur of Maharashtra. Thus answer (b) is correct.

35. The richest state in replenishable groundwater resources is :

(a) Andhra Pradesh (b) Madhya Pradesh
(c) Uttar Pradesh (d) West Bengal

U.P.P.C.S. (Mains) 2006

Ans. (c)

The richest state in replenishable groundwater resources is Uttar Pradesh (2011). According to the status of the year 2024, the annual rechargeable groundwater resources (in bcm) of the State in question are as follows :

State	Replenishable groundwater resource (in billion cubic meter/year)
Andhra Pradesh	27.80
Madhya Pradesh	35.90
Uttar Pradesh	72.84
West Bengal	25.89

Source - National Compilation on Dynamic Ground Water Resources of India, 2024.

36. Which one of the following states is the pioneer in introducing contract farming in India ?

(a) Haryana (b) Punjab
(c) Tamil Nadu (d) Uttar Pradesh

U.P.P.C.S. (Mains) 2005

Ans. (b)

Punjab is the pioneer in introducing contract farming in India.

37. "Green agriculture" involves –

(a) Organic farming and emphasis on horticulture
(b) Avoiding pesticides, chemical fertilizers while focusing on horticulture and floriculture
(c) Integrated pest management, integrated nutrient supply and integrated natural resource management
(d) Emphasis on food crops, horticulture and floriculture

U.P.P.C.S. (Mains) 2016

Ans. (c)

Green Agriculture involves the adoption of integrated pest management, integrated nutrient supply, scientific water management and use of appropriate crop varieties.

38. Which of the following is not true in respect of globalization impact on Indian Agriculture?

(a) Climate change
(b) Emphasis on cash crops

(c) Growth of income inequality
(d) Reduction of subsidies

Uttarakhand Lower Sub. (Pre) 2010

Ans. (a)

Narasimha Rao Government made major modifications in its economic policy in 1991 by introducing Globalization in India. It had a large impact on Indian Agriculture which includes an emphasis on cash crops, growth of income inequality, reduction of subsidies and many more. Hence, option (a) is not true in respect of globalization impact on Indian agriculture.

39. Which one of the following best describes the main objective of 'Seed Village Concept'?

(a) Encouraging the farmers to use their own farm seeds and discouraging them to buy the seeds from others
(b) Involving the farmers for training in quality seed production and thereby to make available quality seeds to others at the appropriate time and affordable cost.
(c) Earmarking some villages exclusively for the production of certified seeds
(d) Identifying the entrepreneurs in villages and providing them technology and finance to set up seed companies.

I.A.S. (Pre) 2015

Ans. (b)

Seed village concept refers to a village wherein trained group of farmers are involved in the production of seeds of various crop and cater to their need of seeds themselves, fellow farmers of their village and the neighbouring village at affordable cost and at the appropriate time". The concept is to increase the seed production, meet local demands at a reasonable cost, self-sufficiency and self-reliance by a trained group of farmers.

40. Agmark is –

(a) Co-operative Committee for production of eggs
(b) Co-operative committee for farmers
(c) Regulated market of eggs
(d) Mark of standard Guarantee (Quality Certification)

U.P.P.C.S. (Pre) 2003

Ans. (d)

Agmark is a quality certification mark provided by the Government of India which is under Agricultural produce (Grading and Marking) Act, 1937. This Act provides for the grading and marking of agricultural and other produce. This plan has been implemented in India since 1938.

41. The size of marginal landholding in India is :

(a) more than 5 hectares
(b) 2 hectares to 4 hectares
(c) 1 hectare to 2 hectares
(d) less than 1 hectare
(e) None of the above/More than one of the above.

63rd B.P.S.C. (Pre) 2017

Ans. (d)

According to the Agriculture Census 2015-16, the number of operational holding and the size of the are group are as follows-

Size group	**Shape**
Marginal landholdings	less than 1 hectare
Small holding	1 - 2 hectares
Semi - medium holding	2 - 4 hectares
Medium holding	4 - 10 hectares
Big holding	More than 10 hectares

42. With reference to the circumstances in Indian agriculture, the concept of "Conservation Agriculture" assumes significance. Which of the following fall under the Conservation Agriculture?

1. Avoiding monoculture practices
2. Adopting minimum tillage
3. Avoiding the cultivation of plantation crops.
4. Using crop residues to cover soil surface
5. Adopting spatial and temporal crop sequencing/ crop rotations.

Select the correct answer using the code given below:

(a) 1, 3 and 4
(b) 2, 3, 4 and 5
(c) 2, 4 and 5
(d) 1, 2, 3 and 5

I.A.S. (Pre) 2018

Ans. (c)

Conservation agriculture is a method that keep agricultural cost low. High profits and sustainable product can be brought. Along with this, there is a balanced increase in natural resources like soil, water, environment, and biological factors. In this, there is a balanced and proper use of agricultural chemicals and inorganic and organic sources along with agricultural activities such as zero tillage/minimum tillage, so that there is no adverse effect on the various bio-functions of agriculture. In conservation agriculture, minimum tillage allows crop residues to remain on the soil surface. This greatly reduces soil erosion. Generally it is very important to keep the soil covered by crop residues up to 30 percent. It is very important to adopt crop diversification and crop rotation in protected cultivation.Generally, farmers follow the same type of crop rotation for many years. For example, farmers have been planting the paddy wheat cropping system in the same field for years, which directly affected the fertility of the soil. It does impact crop diversification maintains soil fertility and reduces crop related pests and diseases.